

Math 262 – Solutions to Practice Problems

1. Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be given by

$$T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} x_1 + x_2 \\ x_2 + x_3 \end{bmatrix}.$$

(a) Show that T is a linear transformation.

(b) Find the matrix of T .

Solution.

1.

$$\begin{aligned} a^{(*)} \quad T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}\right) &= T\left(\begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \\ x_3 + y_3 \end{bmatrix}\right) \\ &= \begin{bmatrix} x_1 + y_1 + x_2 + y_2 \\ x_2 + y_2 + x_3 + y_3 \end{bmatrix} \\ &= \begin{bmatrix} x_1 + x_2 \\ x_2 + y_3 \end{bmatrix} - \begin{bmatrix} x_1 + x_2 + y_1 + y_2 \\ x_2 + y_2 + x_3 + y_3 \end{bmatrix} + \begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \\ x_3 + y_3 \end{bmatrix} \\ &= T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) + T\left(\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}\right). \end{aligned}$$

(i) For $r \in \mathbb{R}$:

$$T\left(r \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} rx_1 + rx_2 \\ rx_2 + rx_3 \end{bmatrix} = r \begin{bmatrix} x_1 + x_2 \\ x_2 + x_3 \end{bmatrix} = r T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right).$$

So T is linear.

2.

$$T(e_1) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad T(e_2) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad T(e_3) = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \Rightarrow A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

2. Given $A\vec{x} = \vec{b}$, suppose the rref of the augmented matrix is

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 2 & 1 \\ 0 & 0 & 1 & 3 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right].$$

Give the general solution to the system.

Solution.

$$\begin{cases} x_1 + x_4 = 0 \\ x_2 + 2x_4 = 1 \\ x_3 + 3x_4 = 2 \end{cases} \Rightarrow \begin{cases} x_1 = -x_4 \\ x_2 = 2x_4 + 1 \\ x_3 = -3x_4 + 2 \end{cases} \quad (x_4 = r).$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -r \\ 2r + 1 \\ -3r + 2 \\ r \end{bmatrix}.$$

Problem 3 (Original Statement): Given $\begin{cases} x_1 - x_3 = 1, \\ 2x_1 - 2x_2 = 1, \\ 3x_1 - 4x_2 + x_3 = a, \end{cases}$ find (a) no solution,

(b) exactly one solution, (c) infinitely many solutions; give the general solution when infinite.

Solution.

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ -2 & 2 & 0 & 1 \\ 3 & -4 & 1 & a \end{array} \right]$$

Row ops: $R_2 \rightarrow R_2 + 2R_1$, $R_3 \rightarrow R_3 - 3R_1$.

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 0 & -2 & 2 & 1 \\ 0 & -4 & 4 & a - 3 \end{array} \right]$$

Row op: $R_2 \rightarrow (-\frac{1}{2})R_2$.

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 0 & 1 & -1 & \frac{1}{2} \\ 0 & -4 & 4 & a - 3 \end{array} \right]$$

Row op: $R_3 \rightarrow R_3 + 4R_2$.

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 0 & 1 & -1 & \frac{1}{2} \\ 0 & 0 & 0 & a - \frac{1}{2} \end{array} \right]$$

(a) No solution if $a \neq 1$. (b) Never exactly one. (c) Infinitely many if $a = 1$.

If $a = 1$, let $x_3 = r$. Then $x_2 = \frac{1}{2} + r$, $x_1 = 1 + r$, so $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 + r \\ \frac{1}{2} + r \\ r \end{bmatrix}$.

In $\mathbb{R}^2$, let $\vec{u} = \begin{bmatrix} \frac{1}{\sqrt{10}} \\ \frac{3}{\sqrt{10}} \end{bmatrix}$.

(a) Show that $\vec{u}$ is a unit vector.

(b) Let L be the line spanned by $\vec{u}$, and let $\vec{x} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$. Find the reflection $\text{ref}_L(\vec{x})$.

Solution.

$$|\vec{u}| = \sqrt{\left(\frac{1}{\sqrt{10}}\right)^2 + \left(\frac{3}{\sqrt{10}}\right)^2} = 1, \quad \text{proj}_L(\vec{x}) = \frac{\vec{x} \cdot \vec{u}}{\vec{u} \cdot \vec{u}} \vec{u}.$$

$$\vec{x} \cdot \vec{u} = \begin{bmatrix} -1 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} \frac{1}{\sqrt{10}} \\ \frac{3}{\sqrt{10}} \end{bmatrix} = -\frac{1}{\sqrt{10}} + \frac{12}{\sqrt{10}} = \frac{11}{\sqrt{10}}, \quad \text{proj}_L(\vec{x}) = \begin{bmatrix} \frac{11}{10} \\ \frac{33}{10} \end{bmatrix}.$$

$$\text{ref}_L(\vec{x}) = 2 \text{proj}_L(\vec{x}) - \vec{x}.$$

Original Problem: In $\mathbb{R}^3$, let L be the line spanned by $\vec{w} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ 1 \end{bmatrix}$, $\vec{x} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$. Verify that

$\vec{x} - \text{proj}_L(\vec{x})$ is orthogonal to $\vec{w}$.

Solution.

$$2 \begin{bmatrix} \frac{11}{10} \\ \frac{33}{10} \\ \frac{11}{10} \end{bmatrix} - \begin{bmatrix} -1 \\ 4 \end{bmatrix} = \begin{bmatrix} \frac{22}{10} \\ \frac{66}{10} \\ \frac{22}{10} \end{bmatrix} - \begin{bmatrix} -1 \\ 4 \end{bmatrix} = \begin{bmatrix} \frac{22}{10} + 1 \\ \frac{66}{10} - 4 \\ \frac{22}{10} \end{bmatrix} = \begin{bmatrix} \frac{16}{5} \\ \frac{13}{5} \\ \frac{12}{5} \end{bmatrix}.$$

Note: matrix formula also works.

$$\text{proj}_L(\vec{x}) = \frac{\vec{x} \cdot \vec{w}}{\vec{w} \cdot \vec{w}} \vec{w} = \frac{2}{\sqrt{2}} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

$$\vec{x} - \text{proj}_L(\vec{x}) = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ 1 \end{bmatrix} = 0.$$

Problem 6 (Original Statement): Define $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by $T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_1 - x_2 \\ x_1 + x_2 \end{bmatrix}$. Assume T is linear.

(a) Find A , the matrix of T .

(b) Use A to explain why T is invertible.

(c) Use A to find the rule for T^{-1} .

Solution.

$$T(e_1) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad T(e_2) = \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \quad A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}.$$

$$\det A = (1)(1) - (-1)(1) = 2 \Rightarrow A \text{ invertible}, \quad A^{-1} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}, \quad T^{-1}(\vec{x}) = A^{-1}\vec{x}.$$

Problem 7 (Original Statement): Let A be rotation by θ . Show $|A\vec{u}| = |\vec{u}|$ for $\vec{u} = \begin{bmatrix} x \\ y \end{bmatrix}$.

Solution.

$$A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}, \quad A\vec{u} = \begin{bmatrix} x \cos \theta - y \sin \theta \\ x \sin \theta + y \cos \theta \end{bmatrix}.$$

$$|A\vec{u}| = \sqrt{(x \cos \theta - y \sin \theta)^2 + (x \sin \theta + y \cos \theta)^2} = \sqrt{x^2(\cos^2 + \sin^2) + y^2(\sin^2 + \cos^2)} = |\vec{u}|.$$

Problem 8 (Original Statement): Fill in the blanks for an invertible $n \times n$ matrix A : (a) $\text{rank}(A) =$ _____ (b) $A\vec{x} = \vec{0}$ has only the _____ solution (c) A row-reduces to the _____ matrix.

Solution. $\text{rank}(A) = n$, trivial solution only, identity matrix.

Let $A = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 \\ 2 & 3 & 4 & 0 & 1 \\ 3 & 4 & 0 & 1 & 5 \\ 6 & 8 & 4 & 2 & 6 \end{bmatrix}$. Give the reduced row echelon form of A .

Solution.

$$\begin{bmatrix} 1 & 1 & 0 & 1 & 0 \\ 2 & 3 & 4 & 0 & 1 \\ 3 & 4 & 0 & 1 & 5 \\ 6 & 8 & 4 & 2 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 4 & -2 & 1 \\ 0 & 1 & 0 & -2 & 5 \\ 0 & 2 & 4 & -4 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 4 & -2 & 1 \\ 0 & 0 & -4 & 0 & 4 \\ 0 & 0 & -4 & 0 & 4 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 4 & -2 & 1 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & -5 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Problem 10(a) (Original Statement): Let $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$. If $AB = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$, show $B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.

Solution. $B = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \Rightarrow AB = \begin{bmatrix} a + 2c & b + 2d \\ 2a + c & 2b + d \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$. So $a + 2c = 2a + c = 0 \Rightarrow a = c = 0$ and $b + 2d = 2b + d = 0 \Rightarrow b = d = 0$.

Problem 10(b) (Original Statement): Let $C = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$. Find all D with $CD = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.

Solution. $D = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \Rightarrow CD = \begin{bmatrix} c & d \\ c & d \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$. Hence $c = d = 0$ and $D = \begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix}$.

Problem 11 (Original Statement): In $\mathbb{R}^2$, let $\vec{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\vec{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. Show any $\vec{w} = \begin{bmatrix} a \\ b \end{bmatrix}$ can be written $\vec{w} = r\vec{u} + s\vec{v}$.

Solution. Solve $r + 2s = a$, $2r + s = b \Rightarrow s = (2a - b)/3$, $r = (-a + 2b)/3$. Thus

$$\begin{bmatrix} a \\ b \end{bmatrix} = \frac{-a+2b}{3} \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \frac{2a-b}{3} \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

(a) Is $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ invertible?

(b) Let $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$. Find A^{-1} and solve $A\vec{x} = \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix}$.

Solution. (a) Row-reduction gives a zero row $\Rightarrow$ not invertible. (b) Row-reducing $[A|I]$ yields $A^{-1} = \begin{bmatrix} 1 & 0 & -1 \\ -2 & 1 & 2 \\ -2 & -1 & -1 \end{bmatrix}$, so $\vec{x} = A^{-1} \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 3 \end{bmatrix}$.

Problem 13 (Original Statement): Let $A = \begin{bmatrix} 1 & r \\ 2r & 1 \end{bmatrix}$.

(a) $\det A$.

(b) For which r is A invertible?

(c) When invertible, give A^{-1} .

Solution. $\det A = 1 - 2r^2$. A invertible for $r \neq \pm \frac{1}{\sqrt{2}}$. $A^{-1} = (1/(1 - 2r^2)) \begin{bmatrix} 1 & -r \\ -2r & 1 \end{bmatrix}$.

Quick Reference – Matrix Ops, REF/RREF, Rank/Det, Solving $A\vec{x} = \vec{b}$

Matrix ops. $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$. $A + B = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}$, $A - B = \begin{bmatrix} -4 & -4 \\ -4 & -4 \end{bmatrix}$, $2A = \begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix}$.

Row $\times$ column: $AB = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$. Size rule: $(m \times n)(n \times p) \rightarrow (m \times p)$.

REF vs RREF.

$$\text{REF} \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & 1 & 4 & 5 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \text{RREF} \begin{bmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

REF: zeros below pivots. RREF: pivots are 1 and the only nonzero in their columns.

Rank/Nullity/Invertibility. $\text{rank}(A) = \#$ pivots, nullity $= n - \text{rank}(A)$. $\text{rank}([A|\vec{b}]) > \text{rank}(A) \Rightarrow$ no solution. If consistent: free vars $\Rightarrow$ infinitely many; none $\Rightarrow$ unique. Square: $\det(A) \neq 0 \Leftrightarrow \text{rank}(A) = n \Leftrightarrow \text{rref}(A) = I_n$.

Solving $A\vec{x} = \vec{b}$. 1) Form $[A|\vec{b}]$. 2) Row-reduce to RREF. 3) Pivots = solved vars; nonpivots = parameters. Write $\vec{x} = \vec{x}_p + \vec{x}_h$ (particular + homogeneous). Rank test: if inconsistent row $[0 \dots 0 | c \neq 0]$ appears, no solution.

Determinant essentials. Row swap: sign flip; scale row by k : $\det \times k$; add multiple: $\det$ unchanged. Triangular U : $\det(U) = \prod u_{ii}$. 2×2 : $\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$.